An Efficient and Lightweight Spectral–Spatial Feature Graph Contrastive Learning Framework for Hyperspectral Image Clustering

Aitao Yang, Min Li, Yao Ding, Xiongwu Xiao, *Member, IEEE*, and Yujie He

***Abstract*— Due to the scarcity of prior information and the high complexity of spectral data, hyperspectral image (HSI) clustering presents a significant challenge. Although recent deep clustering methods have demonstrated remarkable performance, their intricate network structures and poor robustness hinder their practical application. To address this issue, we propose an efficient and lightweight spectral–spatial feature graph contrastive learning (S^2GCL) framework for robust HSI clustering. Specifically, we have designed a novel spectral–spatial feature encoder that fully leverages the information in HSI by incorporating both spatial structure and spectral similarity matrices. To establish a lightweight model, we implement several effective designs: First, S^2GCL eliminates the commonly used data augmentation and discriminator in GCL during the generation of positive embeddings. Second, we use a multilayer perceptron (MLP) to produce low-dimensional embeddings instead of relying on graph convolutional networks (GCNs). Third, negative embeddings are generated through row-shuffling, avoiding the use of neural networks. Finally, we propose a multiple boundary loss function to extract complementary information from spatial structures and neighboring nodes, while also constraining the interclass differences between positive and negative examples. We conducted extensive experiments on four publicly available datasets and compared S^2GCL with state-of-the-art clustering methods. The results indicate that S^2GCL achieves satisfactory performance. The code for S^2GCL will be released at https://github.com/ahappyyang/S2GCL.**

***Index Terms*— Graph contrastive learning, graph neural network, hyperspectral image (HSI) clustering, lightweight.**

I. Introduction

HYPERSPECTRAL images (HSIs) are 3-D data captured from specific scenes, consisting of one spectral dimension and two spatial dimensions. Compared with visible images, HSI has hundreds of spectral bands and offers high resolution, allowing for more precise differentiation of land covers. Due to its unique properties, HSI has garnered increasing attention and is widely applied in various fields such as precision agriculture, environmental monitoring [1], object detection [2], land-cover classification [3], military target recognition [4], and resources' prospecting.

HSI classification aims to identify land-cover types accurately by extracting spectral–spatial features. It is also the foundation for precise interpretation and quantitative analysis of hyperspectral data. In the past few decades, a large number of feature extraction methods have emerged for HSI classification. In the early stages, machine learning methods were used to distinguish different types of land cover based on spectral differences, such as random forests [5], Markov random fields [6], and support vector machines (SVMs) [7]. Recently, some deep learning methods have been successfully and widely applied for HSI classification, including convolutional neural networks (CNNs) [8], graph convolutional networks (GCNs) [9], transformer [10], etc. However, most methods are supervised or semisupervised, and manually annotated samples will considerably consume human resources. Unlike supervised or semisupervised methods, HSI clustering does not require annotated samples or relies only on a small amount of prior knowledge. It establishes dependency relationships between land covers and divides land covers into multiple clusters based on the similarity of features. Therefore, HSI clustering has received widespread attention. Due to the imbalance between the unlabeled samples and the rich spatial and spectral details in HSI, high-precision HSI clustering becomes a challenging task [11].

Scholars have explored many different clustering methods to achieve precise and effective clustering. In the early stages, some traditional methods were widely applied, such as K-means [12], spectral clustering (SC) [13], fuzzy C-means (FCM) [14], and independent component analysis (ICA) models. However, these methods often focus solely on the spectral information of pixels, neglecting the rich spatial information present in HSIs, which limits their clustering performance. In recent years, with the development of deep learning, many well-known networks have been applied to HSI clustering, such as autoencoder (AE)-based clustering, separation network-based clustering, generative network-based clustering, and graph-based clustering methods. These methods represent the current mainstream clustering approaches, which use deep neural networks, fully connected networks, and CNNs to learn high-level discriminative features and achieve superior clustering performance.

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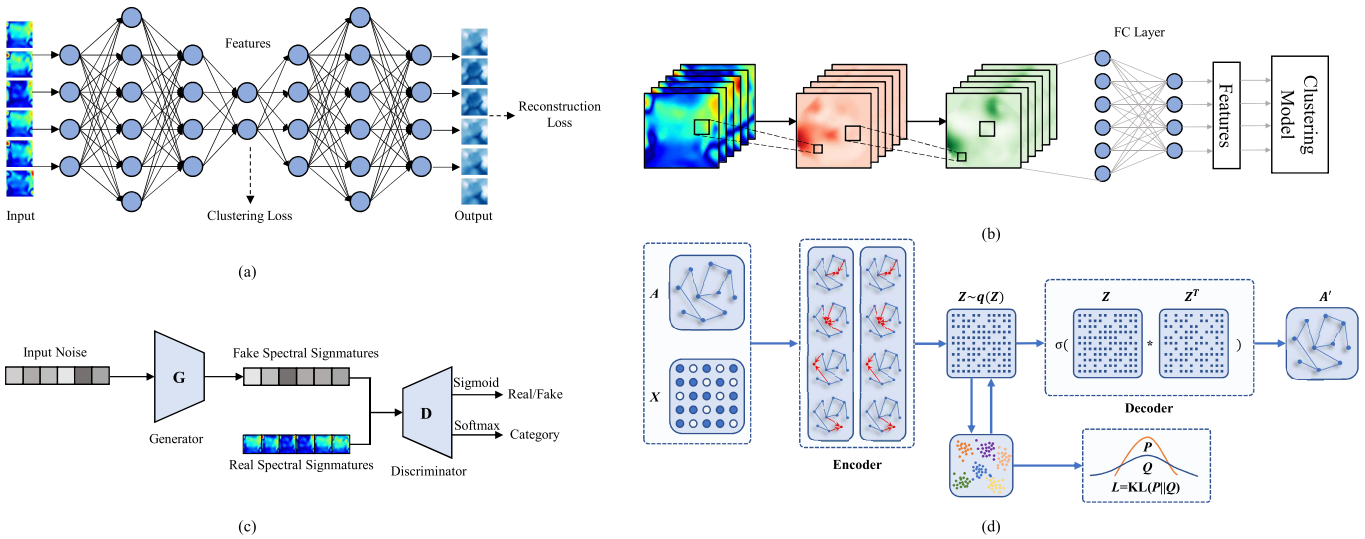


Fig. 1. Overview of currently recognized HSI clustering models, such as (a) AE, (b) separated network, (c) generative network, and (d) graph clustering network.

Autoencoder (AE) [15] typically consists of an encoder for dimensionality reduction and feature extraction and a decoder for data reconstruction. AEs achieve unsupervised high-level semantic information extraction by minimizing reconstruction error, which is simplicity and efficiency. For example, to obtain the nonlinear latent feature representation, Yang et al. [16] proposed a deep clustering network that used an AE to map high-dimensional data to the latent space. It achieved efficient clustering by combining dimensionality reduction features with K-means. In separated-network-based clustering methods, the feature extraction and clustering processes are conducted separately, and the network parameters are optimized through clustering loss. For example, inspired by spectral graph theory and deep learning, Zhao and Li [17] proposed the deep SC with regularized linear embedding (DSCRLE) network. DSCRLE uses a fully connected network to extract latent feature representations, followed by normalization through spectral orthogonal constraints, which enhances the generalizability and scalability of SC in HSI clustering. Generative-network-based clustering methods can not only obtain a more accurate distribution of learning samples through the adversarial interaction between generators and discriminators but also generate new samples through generators to assist decision-making [18]. For example, Zhan et al. [19] achieved high-level semantic information extraction by introducing AEs into adversarial networks, while using the tunable Gaussian mixture prior to further improving classification performance. Compared with extracting features first and then clustering, graph deep clustering can simultaneously learn node representations through graph neural networks while enhancing the cohesion of graph nodes through self-supervised graph clustering. For example, to accommodate higher spectral dimensions and establish spatial relationships between land covers, Wang et al. [20] embedded the spectral features into the combined adjacent graph and proposed a method called spectral–spatial clustering with anchor graph (SSCAG) for HSI clustering. To capture the spatial relationships between different nodes, Wang et al. [21] used a fusion attention mechanism to better aggregate neighbor information for updating node representations. Moreover, it used the inner product of node pairs to reconstruct the original graph structure.

The aforementioned deep clustering networks can achieve nonlinear feature modeling. However, they struggle to fully exploit the relationships between HSI spectra. Moreover, their ability to comprehensively mine low-level structural information and high-level semantic features remains insufficient. Fig. 1 shows the four types of deep clustering networks. Specifically, the above methods have the following limitations.

1) AEs can obtain the data's nonlinear and low-dimensional features by reconstructing the original data. However, AEs can be prone to overfitting, especially when the network structure is complex or the amount of training data is insufficient.
2) The generative networks are prone to collapse in training due to the class imbalance of land-cover types, causing great uncertainty and affecting clustering performance.
3) Due to the high spectral variability and dimensionality of HSI, graph-clustering-based methods may encounter the "curse of dimensionality" issue when dealing with high-dimensional data, resulting in poor clustering performance.

The drawbacks of the above methods limit further improvement of HSI clustering accuracy. In recent years, graph contrastive learning (GCL) has been gradually applied in computer vision, natural language processing, and other fields due to its powerful ability to mine the structure of data [22], [23], [24]. Graph contrastive learning achieves low-level structural information fusion and high-level semantic feature extraction through graph structure embedding and data augmentation. It provides self-supervised signals for the model through contrastive learning of positive and negative embeddings, making model optimization simpler and generalization performance stronger. For example, You et al. [25] explored the effectiveness of various data augmentation methods on different

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datasets and proposed a new graph contrastive learning framework (GraphCL) to enhance the transferability and robustness of GCL. To better use the similarity of nodes for feature extraction, Chen and Kou [26] developed a contrastive learning framework that preserved attributes and structures in graphs and used three different graph views for contrastive learning to mine more comprehensive graph information. To solidify the hierarchical substructures in the graph, DualGCL [27] was proposed for self-supervised graph-level representation learning, using a differentiable transformer aggregator for adaptive hierarchical aggregation to achieve fine-grained local information merging.

HSI contains rich spatial information and high-resolution spectra so a robust feature representation network is required for HSI clustering. Therefore, this article rethought the feature extraction paradigm of HSIs, and the GCL method was introduced to explore the ability of feature mining in HSI clustering. However, some challenges should be overcome for HSI clustering.

1) The effectiveness of GCL feature extraction largely depends on data augmentation. However, due to the diversity of graph structures, it is challenging to effectively use various attributes, structural features, and semantic relationships in graph data for data augmentation, which significantly impacts the efficiency and generalizability of the model [28].
2) GCL typically requires the multiple encoders to create different comparison views, as well as calculate the similarity between a large number of positive and negative sample pairs, which greatly increases model complexity and time cost.

To fully leverage the advantages of GCL in HSI clustering, we propose an efficient and clustering framework of graph contrastive learning with lightweight spectral–spatial feature embedding, spectral–spatial feature GCL (S^2GCL), which can effectively improve clustering performance and reduce model complexity. Specifically, simple linear iterative clustering (SLIC) is used to segment similar features to obtain superpixels, as well as establish graph structures. Subsequently, a multilayer perceptron (MLP) is used to generate anchor embeddings. Afterward, the spectral–spatial graph convolutional encoder (S^2GCN) and the lightweight neighbor encoder are designed for generating structural complementary positive embeddings. Negative embeddings are generated by random row-shuffling. Finally, multiple boundary loss is designed to enhance the compactness of anchor embeddings with positive embeddings and the separability of anchor embeddings from negative embeddings by introducing boundary distances.

Compared with previous methods, the innovative contributions of S^2GCL are as follows.

1) The S^2GCN encoder is proposed, which introduces a spatial structure matrix and a spectral similarity matrix for embedding, while integrating a lightweight neighbor encoder to facilitate the joint extraction of structural and neighboring information.
2) S^2GCL abandons the previously widely used data augmentation and discriminators, instead generating negative embeddings through row-shuffling, which reduces model complexity and improves computational efficiency.
3) Multiple boundary loss is designed to further constrain the distances among anchor embeddings, positive embeddings, and negative embeddings using boundary distance.

The remainder of this article is organized as follows: Section II introduces the relevant theories of contrastive learning and GCNs. Section III first introduces the overview of S^2GCL, and then elaborates on important structural details such as the carefully designed S^2GCN encoder and multiple boundary loss. Section IV conducts extensive experiments and result analysis. Section V provides a comprehensive conclusion and future research direction.

II. Review of Contrastive Learning and GCN

A. Contrastive Learning

To reduce the excessive reliance on annotated samples during the training, contrastive learning mines supervisory information from the internal structure of a large amount of unlabeled data. In addition, contrastive learning could extract transferable knowledge or features through self-supervised learning for model training.

Assuming x is a sample, the purpose of contrastive learning is to learn a mapping function $f(\cdot)$ that satisfies the following equation:

$$s\left(f(x), f\left(x^{+}\right)\right) >> s\left(f(x), f\left(x^{-}\right)\right) \tag{1}$$

where x^+ is the positive sample that is similar to x, x^- is the negative sample with differences from x, and $s(\cdot)$ is a function that measures the similarity between samples.

The foundation of feature extraction in contrastive learning is to construct positive and negative sample pairs reasonably and efficiently and to maximize the semantic information covered by sample pairs. Contrastive learning reduces the distance between anchors and positive samples and increases the distance between anchors and negative samples by comparing the loss function. The constraints imposed by the loss function make the similarity between positive samples and anchors much more significant than between negative samples and anchors, thereby improving the model's ability to extract sample information.

B. Graph Convolutional Network

GCNs can represent data as undirected graphs, thereby establishing relationships among different land-cover types. By establishing spatial dependencies for these land covers, GCN further enhances the model's classification performance. For a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, $\mathcal{V}$ represents a set of vertices, $\mathcal{E}$ is the edge set, and $\mathbf{A} \in \mathrm{R}^{N\times N}$ is the adjacency matrix of $\mathcal{G}$. Given the adjacency matrix $\mathbf{A}$, the Laplace matrix $\mathbf{L}$ of the $\mathcal{G}$ is

$$\mathbf{L} = \mathbf{D} - \mathbf{A} \tag{2}$$

where $\mathbf{D} \in \mathrm{R}^{N\times N}$ is the degree matrix of $\mathcal{G}$.

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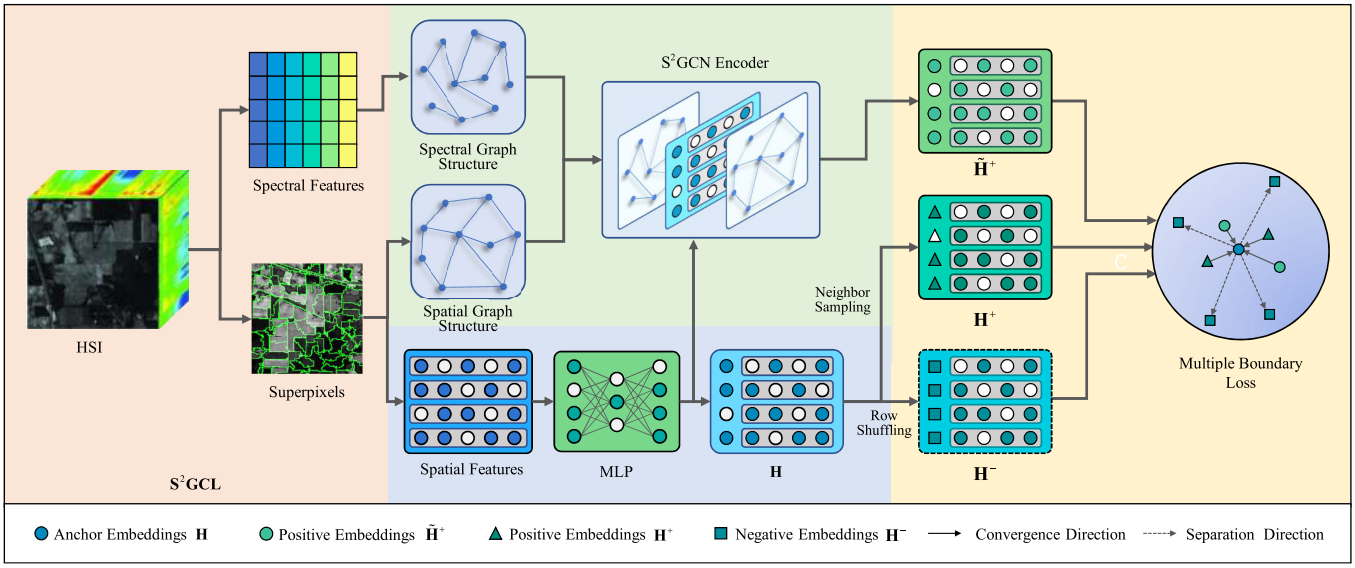


Fig. 2. Overall flowchart of S^{2}GCL. Specifically, homogeneous regions are formed through superpixel segmentation; spatial semantic features are transformed into anchor embeddings through MLP. The S^{2}GCN encoder introduces the spatial structure matrix and spectral similarity matrix to generate positive embeddings with spectral–spatial information. Moreover, neighboring positive embeddings are obtained by aggregating domain information. Furthermore, negative embeddings are generated through row-shuffling. Finally, multiple boundary loss is designed to learn potential features.

Normalizing the Laplace matrix $\mathbf{L}$ yields

$$\mathbf{L}_n = \mathbf{I} - \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}} \tag{3}$$

where $\mathbf{I} \in \mathrm{R}^{N\times N}$ is the identity matrix.

GCN introduces a convolutional kernel $g_\theta = \mathrm{diag}(\theta)$ to embed nodes, where $\theta \in R^N$. The formula for graph convolution can be expressed as

$$x_G * g_\theta = \mathbf{U} g_\theta(\Lambda)\mathbf{U}^T x \tag{4}$$

where $x_G \in \mathrm{R}^{N\times D}$ is the node representation, $\mathbf{U} \in \mathrm{R}^{N\times N}$ is the eigenvector matrix of $\mathbf{L}_n$, and Λ is the eigenvalues of $\mathbf{L}_n$. To simplify the calculation, the Chebyshev polynomial is used to fit $g_\theta(\Lambda)$

$$g_\theta(\Lambda) \approx \sum_{k=0}^{K} \theta_K T_K(\mathbf{L}_n) \tag{5}$$

where T_K is the Chebyshev polynomial.

Taking the first-order Chebyshev polynomial for further simplification, we can obtain

$$x_G * g_\theta = \theta\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}\right)x. \tag{6}$$

Based on (6), the propagation rule of GCN is as follows:

$$\mathbf{H}^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}\mathbf{H}^{(l)}\mathbf{W}^{(l)}\right) \tag{7}$$

where $\mathbf{H}^{(l+1)} \in \mathrm{R}^{N\times D'}$ and $\mathbf{H}^{(l)} \in \mathrm{R}^{N\times D'}$ represent the features of the $l+1$ and l layers respectively, $\sigma(\cdot)$ represents the activation function, and $\mathbf{W}$ represents the weight matrix.

III. Proposed Methodology

In this section, we start to elaborate on the workflow and overall structure of S^{2}GCL. Subsequently, we provide a detailed introduction to anchor embedding and positive embedding of joint spectral–spatial structure information. Moreover, the lightweight neighbor embedding and shuffle negative embedding are elaborated. Finally, the construction process of multiple boundary loss is discussed.

A. Overview of S^{2}GCL

Our motivation is to design a new lightweight and universal efficient clustering network based on GCL, which focuses on the rich spatial information and high-resolution spectral information of HSI.

From the overview in Fig. 2, we can divide our method into three parts: superpixel graph structure construction, lightweight encoder, and multiple boundary loss.

1) *Superpixel Graph Structure Construction:* The superpixel segmentation algorithm divides land cover with similar features into the same area, reducing the number of nodes. In addition, S^{2}GCL uses the average spectral value as the node feature, saving computational resources and further obtaining the graph representation.
2) *Lightweight Encoder:* S^{2}GCL abandons data augmentation and discriminators and only uses MLP to generate anchors, making the network parameters lightweight and improving clustering efficiency. Furthermore, the S^{2}GCN encoder is designed to integrate HSI spectral–spatial information and it combines lightweight negative embedding to achieve abstract semantic information extraction.
3) *Multiple Boundary Loss:* By introducing boundary distance, the anchor embeddings are closer to positive embeddings and farther away from negative embeddings.

The above three components are embedded into the same clustering network, and each component benefits the other components, forming an end-to-end lightweight graph contrastive learning clustering network S^{2}GCL.

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B. Anchor Embedding and S^{2}GCN Encoder

In graph contrastive learning, anchor embeddings are often obtained by the GCN encoder. For example, neighbor contrastive learning on leachable graph augmentation (NCLA) [29] used a multihead graph attention mechanism, and an independent graph encoder was used to generate anchor embeddings. Chen and Kou [26] developed a structure-preserving graph contrastive learning framework (ASP), which generated three views and obtained anchor embeddings through the GCN encoder. However, in the process of generating anchors through GCN, the calculation of the adjacency matrix and forward propagation process leads to model parameters being numerous and time-consuming.

To improve computational efficiency and model universality, this article uses MLP instead of GCN to generate anchor embeddings $\mathbf{H}$ with semantic information for input $\mathbf{X}$

$$\begin{aligned}\mathbf{X}^{(l+1)} &= \mathrm{ReLU}\left(\mathbf{X}^{(l)}\mathbf{W}^{(l)}\right) + b^{(l)} \\ \mathbf{H} &= \mathrm{Dropout}\left(\mathbf{X}^{(l+1)}\mathbf{W}^{(l+1)}\right)\end{aligned} \tag{8}$$

where $\mathbf{X}^{(l)} \in \mathrm{R}^{N\times D}$, $\mathbf{W}^{(l)} \in \mathrm{R}^{D\times D}$, and $b^{(l)} \in \mathrm{R}^{N\times D}$ represent the output features, weights, and biases of the lth layer, respectively. ReLU$(\cdot)$ represents the activation function. Dropout$(\cdot)$ is used to improve the model's generalization ability.

Generally speaking, the positive embeddings are obtained through an encoder with spatial graph structural information [30]. However, their rich and almost continuous spectral information is the key to achieving accurate clustering for HSIs. Unlike previous methods, this article designs the S^{2}GCN encoder, which introduces the spatial structure of the graph while fully using spectral domain information to construct positive embeddings.

The S^{2}GCN encoder characterizes the spectral–spatial relationship of HSI by constructing the spatial structure matrix and the spectral similarity matrix.

Spatial structure matrix: Let the set of k nearest neighbors of the ith and jth superpixels $\mathbf{p}_i$ and $\mathbf{p}_j$ be $N(\mathbf{p}_i)$ and $N(\mathbf{p}_j)$, respectively. If $\mathbf{p}_i$ belongs to $N(\mathbf{p}_j)$ or $\mathbf{p}_j$ belongs to $N(\mathbf{p}_i)$, then the relationship between $\mathbf{p}_i$ and $\mathbf{p}_j$ can be calculated using the Gaussian kernel function, otherwise it is 0

$$A_{ij} = \begin{cases} \exp\left(-\frac{\|\mathbf{p}_i-\mathbf{p}_j\|^2}{2\sigma^2}\right), & \mathbf{p}_i \in N\left(\mathbf{p}_j\right) \text{ or } \mathbf{p}_j \in N(\mathbf{p}_i) \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

Therefore, we can obtain the spatial structure matrix $\mathbf{A} \in \mathbb{R}^{N\times N}$, where N is the number of superpixels. Unlike the adjacency matrix representation that uses 0 and 1 to denote adjacency relationships, the spatial structure matrix can better represent spatial connectivity. It not only reflects the adjacency relationships between different graph nodes but also captures the similarity between nodes, enabling the model to focus more effectively on crucial spatial information.

Spectral Similarity Matrix: Let the sets of k nearest neighbors of the mth and nth spectra $\mathbf{q}_m$ and $\mathbf{q}_n$ be $K(\mathbf{q}_m)$ and $K(\mathbf{q}_n)$, respectively. If $\mathbf{q}_m$ belongs to $K(\mathbf{q}_n)$ or $\mathbf{q}_n$ belongs to $K(\mathbf{q}_m)$, then the relationship between $\mathbf{q}_m$ and $\mathbf{q}_n$ must be calculated using the Gaussian kernel function, otherwise it is 0

$$S_{mn} = \begin{cases} \exp\left(-\frac{\|\mathbf{q}_m-\mathbf{q}_n\|^2}{2\sigma^2}\right), & \mathbf{q}_m \in K(\mathbf{q}_n) \text{ or } \mathbf{q}_n \in K(\mathbf{q}_m) \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

Therefore, we can obtain the spectral similarity matrix $\mathbf{S} \in \mathbb{R}^{D\times D}$, where D is the number of spectra. The spectral similarity matrix establishes a local spectral attention mechanism, which makes the connections between spectra more closely related, facilitating the joint clustering of multiband information. The spatial structure matrix emphasizes the connectivity and topological relationships of the graph, while the spectral similarity matrix focuses on the similarities between spectra. These matrices provide a comprehensive representation that enhances the ability of the S^{2}GCN encoder to learn from complex graph structures.

To alleviate the situation of gradient vanishing or exploding, this article regularizes by normalizing $\mathbf{A}$ and $\mathbf{S}$

$$\hat{A} = \tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}} \tag{11}$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}_N$, $\mathbf{I}_N \in \mathbb{R}^{N\times N}$ is the identity matrix, $\tilde{\mathbf{D}} \in \mathbb{R}^{N\times N}$ is the degree matrix of $\tilde{\mathbf{A}}$, and finally it obtains the normalized spatial structure matrix $\hat{A} \in \mathbb{R}^{N\times N}$. Similarly, the normalized spectral similarity matrix $\hat{S} \in \mathbb{R}^{D\times D}$ can be obtained.

The spectral–spatial relationship embeddings of land covers are crucial for feature encoding. Recently, there has been widespread attention on using graph representations to achieve spatial position embeddings [31], [32], [33]. For example, Jiang et al. [34] proposed a new graph generative structure-aware transformer (GraphGST) to replace randomly generated positional encoding, enhancing the model's representation capability. Hong et al. [35] introduced a new minibatch GCN that establishes spatial relationships between pixels using an adjacency matrix and uses three fusion strategies to achieve performance gains.

In the S^{2}GCN encoder, we couple the spatial structure and spectral relationships to enable the encoder to learn discriminative representations of HSI spectral–spatial information better. The S^{2}GCN encoder is constructed as follows:

$$\tilde{\mathbf{H}}^{+(l+1)} = \mathrm{ReLU}\left(\hat{A}\mathbf{H}^{+(l)}\hat{S}\mathbf{W}^{(l)}\right) + \mathbf{b}^{(l)} \tag{12}$$

where $\mathbf{W}^{(l)} \in \mathbb{R}^{D\times d}$ and $\mathbf{b}^{(l)} \in \mathbb{R}^{D\times d}$ represent the weights and biases of the lth layer, respectively. D and d represent the dimensions of the input features and output features, respectively. $\mathbf{H}^{+(l)} \in \mathbb{R}^{N\times D}$ represents the features of the lth layer. $\tilde{\mathbf{H}}^{+(l+1)} \in \mathbb{R}^{N\times d}$ is the spectral–spatial embedding obtained through the S^{2}GCN encoder.

C. Lightweight Neighbor Embedding and Negative Embedding

GCL often obtains new graph views through data augmentation and then encodes the views to obtain positive and negative embeddings. For example, Zhang et al. [36] generated two new views through data augmentation and encoded them through

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a GCN encoder that shared parameters. Xuheng et al. [37] used singular value decomposition for graph expansion and improved the performance of contrastive learning by introducing global structural relationships.

To alleviate the pressure caused by excessive computational load, this article abandons the data augmentation to obtain new views and instead designs neighbor embeddings using graph neighbor information based on anchor embeddings

$$\begin{aligned}\mathbf{h}_i^+ &= \alpha \mathcal{F}(\mathbf{h}_j) + \mathbf{x}_i \\ &= \alpha \times \frac{1}{n} \sum_{j=1}^{n} \{\mathbf{h}_j \mid v_j \in \mathcal{N}_i\} + \mathbf{x}_i \end{aligned} \tag{13}$$

where $\mathcal{F}(\cdot)$ represents the neighbor node sampling function, α is a constant, n represents the number of sampled neighboring nodes, $\mathbf{h}_j \in \mathrm{R}^{1\times D}$ represents the anchor embeddings, and $\mathcal{N}_i$ represents the set of neighboring nodes for node v_i. At the same time, introducing the original node feature $\mathbf{x}_i$ to retain a part of the original node features, resulting in the final neighborhood embedding $\mathbf{h}_i^+ \in \mathrm{R}^{1\times D}$.

Neighbor embedding focuses more on local neighbor information, while spectral–spatial embedding focuses on global graph structure information and spectral relationships. Neighbor and spectral–spatial embedding capture different structural information of the graph for better effective contrastive learning.

GCL usually destroys the graph structure and then encodes the damaged graph structure through GCN to obtain negative embeddings [28]. In S^{2}GCL, negative embeddings are obtained through row-shuffling, which reduces two cumbersome steps to achieve a lightweight model

$$\mathbf{H}^- = \mathrm{Shuffle}([\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_N]) \tag{14}$$

where Shuffle$(\cdot)$ represents row-shuffle anchor embeddings, and $\mathbf{H}^- \in \mathrm{R}^{N\times D}$ is the negative embeddings.

It is worth noting that when designing neighbor and negative embeddings, S^{2}GCL did not use any data augmentation methods or neural networks for encoding. This significantly improves the model's computational efficiency, and the experiment's results in Section IV-D prove that this design is simple and effective.

D. Multiple Boundary Loss

The contrastive learning loss function is often used for self-supervised learning. It forms a self-supervised signal by narrowing the distance between the positive embedding and the anchor and pushing the distance between the negative embedding and the anchor, thereby learning the abstract semantic features. Many contrastive learning methods use discriminators to learn the semantic relationship between positive and negative embeddings, but this approach is very time-consuming. In addition, improving the generalization performance of the model is also crucial for enhancing its adaptability.

Inspired by the above perspective, we design the multiple boundary loss. We abandon the discriminator to improve computational efficiency and introduce boundary distance to improve the robustness of the model.

For each positive and negative embedding pair, the purpose of contrastive learning loss is to make the distance between anchor and positive embeddings smaller than between anchor and negative embeddings. Therefore, it can be expressed as

$$\mathcal{D}(\mathbf{H}, \mathbf{H}^+) + d < \mathcal{D}(\mathbf{H}, \mathbf{H}^-) \tag{15}$$

where $\mathcal{D}(\cdot)$ represents the distance measurement function, which calculates embeddings' similarity. d is a constant greater than 0, representing the boundary distance, to better distance positive and negative samples. The triplet loss representation of (15) is as follows:

$$\mathcal{L} = \max(\mathcal{D}(\mathbf{H}, \mathbf{H}^+) - \mathcal{D}(\mathbf{H}, \mathbf{H}^-) + d, 0) \tag{16}$$

where $\max(\cdot)$ represents the maximum value function. $\mathcal{L}$ is the triplet loss.

In S^{2}GCL, to keep spectral–spatial embeddings and neighboring embeddings as far away as possible from negative embeddings, the following losses are designed:

$$\mathcal{L}_G = \max(\mathcal{D}(\mathbf{H}, \mathbf{H}^+) - \mathcal{D}(\mathbf{H}, \mathbf{H}^-) + d, 0) \tag{17}$$

$$\mathcal{L}_N = \max(\mathcal{D}(\mathbf{H}, \tilde{\mathbf{H}}^+) - \mathcal{D}(\mathbf{H}, \mathbf{H}^-) + d, 0) \tag{18}$$

where $\mathcal{L}_G$ and $\mathcal{L}_N$ represent the spectral–spatial embeddings' loss and neighboring embeddings' loss, respectively.

An efficient GCL should have consistency, meaning the learned positive and anchor embeddings should be as close as possible in the feature space.

Equations (17) and (18) ensure that the anchor embedding is closer to the positive embedding than the negative embedding. However, this can only constrain the relative distance between embeddings. Even if (17) and (18) are nonzero, the distance between anchor embedding and positive embedding may still be significant, which affects the model's generalization performance. Therefore, we improve consistency in GCL by applying boundary constraints to positive and negative embedding pairs

$$\mathcal{D}(\mathbf{H}, \mathbf{H}^+) + d < \mathcal{D}(\mathbf{H}, \mathbf{H}^-) < \mathcal{D}(\mathbf{H}, \mathbf{H}^+) + d + d' \tag{19}$$

where d' is a nonnegative constant, and the upper bound $d+d'$ ensures that the distance between negative embeddings and anchor embeddings is finite. At the same time, based on (15), it also ensures that the distance between anchor embeddings and positive embeddings is finite, improving contrastive learning consistency. The corresponding triplet loss is

$$\mathcal{L}_S = -\min(\mathcal{D}(\mathbf{H}, \mathbf{H}^+) - \mathcal{D}(\mathbf{H}, \mathbf{H}^-) + d + d', 0) \tag{20}$$

where $\mathcal{L}_S$ is the boundary loss.

The final fusion of the above three intermediate losses resulted in the multiple boundary loss as follows:

$$\mathcal{L} = \lambda_1 \mathcal{L}_G + \lambda_2 \mathcal{L}_N + \mathcal{L}_S \tag{21}$$

where λ_1 and λ_2 are the weight coefficients of $\mathcal{L}_G$ and $\mathcal{L}_N$, respectively. Finally, we provide the pseudocode of S^{2}GCL in Algorithm 1.

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Algorithm 1 Mechanism of S^{2}GCL

Input: Input features $\mathbf{X}$; Number of clusters c; Number of epoch $\mathcal{N}$; Loss function weight factors λ_1 and λ_2.

Calculate the Spatial Structure Matrix $\mathbf{A}$ and Spectral Similarity Matrix $\mathbf{S}$ based on (9) and (10), respectively. Normalize the two matrices according to (11).
for n=1 to $\mathcal{N}$ **do**
  Encode the input features $\mathbf{X}$ through MLP to obtain anchor embedding $\mathbf{H}$ according to (8).
  According to (12), $\mathbf{H}$ is fused with the spectral–spatial features through the S^{2}GCN encoder to obtain the spectral–spatial embedding $\tilde{H}$.
  According to the (13), generate the neighborhood embedding $\mathbf{H}^+$ by incorporating the information from neighbor nodes.
  Obtain negative embeddings $\mathbf{H}^-$ using the row-shuffling according to (14).
  Minimize the loss function $\mathcal{L}$ according to (21).
end for
Obtain clustering results using the K-means algorithm.

Output: Predict land-cover classes Y.

E. Computational Complexity Analysis

The MLP and the GCN for a superpixel graph with n nodes and d feature dimensions mainly determine the computational complexity of S^{2}GCL. Assuming l is the number of layers in MLP, the computational complexity of MLP is $\mathcal{O}(ndd_1 + n\sum_{i=2}^{l-1} d_i d_{i-1})$, and the computational complexity of graph convolution module is $\mathcal{O}(n^2 d + nd^2)$. As can be seen, due to the lack of graph expansion modules and discriminators in S^{2}GCL, the model is concise and efficient, with less computational complexity compared with other computational methods.

IV. Experiment

In this section, four well-known HSI datasets and experimental settings are first described. Moreover, the effectiveness of S^{2}GCL is demonstrated by comparing with three classic methods and eight state-of-the-art deep learning models. Finally, the network structure's effectiveness and hyperparameter settings' impact on the experimental results are analyzed through ablation experiments.

A. Data Description

The Salina (SA) dataset is captured by 224 band AVIRIS sensors over Salinas Valley, California, with high spatial resolution features. The coverage area includes 512 × 217 samples. It contains 224 bands, leaving 204 bands after removing interference bands. It includes 16 types of features, with a total of 111 104 pixels, of which 56 975 are background pixels and 54 129 are classified pixels.

The Pavia University (UP) dataset is a portion of hyperspectral data from images taken by Germany's airborne Reflective Optics Spectral Imaging System (ROSIS-03) in 2003 in Pavia, Italy. The spectral imager continuously images 115 bands within the wavelength range of 0.43–0.86 mum, and the spatial resolution of the resulting image is 1.3 m. Among them, 12 bands are removed due to the influence of noise, so the image composed of the remaining 103 spectral bands is generally used. The data size is 610 × 340, so it contains 2 207 400 pixels. The total number of pixels containing features is only 42 776, including nine feature types.

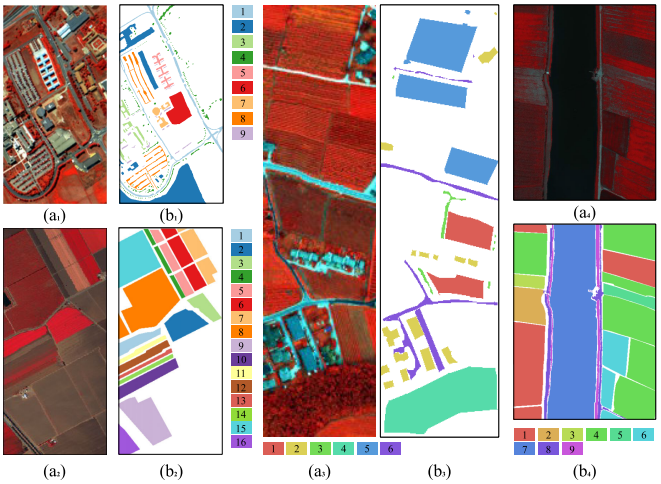


Fig. 3. Details of the four datasets. (a_1)–(a_4) False-color maps of the UP, SA, Trento, and LongKou datasets, respectively. (b_1)–(b_4) Ground truth, and the color blocks correspond to the ground color.

The Trento dataset is captured in a rural area in the southern part of Trento, Italy. It is collected by the AISA Eagle sensor, which consists of 600 × 166 pixels, 63 spectral channels, and a range of 0.40–0.98 μm. The spatial resolution of Trento is 1 m.

The WHU-Hi-LongKou (LongKou) dataset is collected using the DJI Matrice 600 Pro (DJI M600 Pro) UAV platform in Longkou Town, Hubei Province, China. The data collection area is a simple agricultural scene that includes nine types of land cover. The data size is 550 × 400 pixels, with 270 bands and a spatial resolution of 0.463 m.

The specific land covers and the corresponding quantities of the four datasets are shown in Table I, and the ground truth is shown in Fig. 3.

B. Evaluation Indicators

1) Experimental Settings: Overall classification accuracy (OA), kappa coefficient(κ), normalized multiple information (NMI), adjusted rand index (ARI), and Purity are used as quantitative and qualitative evaluation indicators. Among them, OA represents the overall classification accuracy. κ is an indicator used to evaluate the degree of consistency in judgment. NMI measures the similarity between two clustering results. ARI represents the degree of agreement between the results and the real situation. Purity is a simple and transparent evaluation metric. Purity$\in$ [0, 1], and the closer it is to 1, the better the clustering result.

2) Comparison to Other Baselines: To comprehensively analyze the effectiveness of the S^{2}GCL, this article shows comparison of S^{2}GCL with 11 selected baselines, including K-means, FCM, positive C-means (PCMs), NCSC [38], DFCN [39], SDCN [40], EGAE [41], AdaGAE [42],

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TABLE I
DETAILS OF EACH LAND-COVER CLASS OF FOUR DATASETS USED FOR EVALUATION IN EXPERIMENTS

NO.	Salinas		Pavia University		Trento		LongKou	
	Class Name	Training	Class Name	Training	Class Name	Training	Class Name	Training
1	Weeds-1	2009	Asphalt	6631	Apple trees	4034	Corn	34511
2	Weeds-2	3726	Meadows	18649	Buildings	2903	Cotton	8374
3	Fallow	1976	Gravel	2099	Ground	479	Sesame	3031
4	Fallow-rough-plow	1394	Trees	3064	Woods	9123	Broad-leaf soybean	63212
5	Fallow-smooth	2678	Painted metal sheets	1345	Vineyard	10501	Narrow-leaf soybean	4151
6	Stubble	3959	Bare Soil	5029	Roads	3174	Rice	11854
7	Celery	3579	Bitumen	1330			Water	67056
8	Grapes-untrained	11271	Self-Blocking Bricks	3682			Roads and houses	7124
9	Soil-vinyard-develop	6203	Shadows	947			Mixed weed	5229
10	Corn	3278						
11	Lettuce-4wk	1068						
12	Lettuce-5wk	1927						
13	Lettuce-6wk	916						
14	Lettuce-7wk	1070						
15	Vinyard-untrained	7268						
16	Vinyard-trellis	1807						
Total		54129		42776		30214		204542

TABLE II
QUANTITATIVE COMPARISON OF THE SA DATASET, WITH BEST PERFORMANCE IN BOLD

Class No.	K-means	FCM	PCM	NCSC	DFCN	SDCN	EAGE	AdaGAE	DAEGC	SEDESC	DSCRLE	S^2GCL
1	0.9925	0.9851	0.6327	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000	0.0000	0.9557	**1.0000**
2	0.6286	0.7354	0.8070	1.0000	0.9990	0.9994	1.0000	0.9987	0.9997	1.0000	0.9946	**1.0000**
3	0.0000	0.0000	0.0020	0.6538	1.0000	0.3360	1.0000	1.0000	0.0000	0.0000	0.0000	**1.0000**
4	**0.9900**	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.9505	0.0000
5	0.9731	0.9328	0.9776	**0.9985**	0.7308	0.9870	0.0000	0.9642	0.6502	0.9817	0.9698	0.9948
6	0.5168	0.9608	0.9917	0.9970	0.9986	0.9450	0.9917	0.9881	1.0000	1.0000	0.9929	**1.0000**
7	0.6052	0.5742	0.9925	**1.0000**	0.9561	0.0000	0.9979	0.9997	0.9718	0.9958	0.9849	0.9704
8	0.6359	0.2686	0.3934	0.7892	0.4320	0.7633	0.4854	0.5861	0.6683	**0.9946**	0.7015	0.7082
9	0.9421	0.9325	0.5702	0.9183	0.9063	0.8750	0.8446	0.9752	0.9141	1.0000	0.9924	**1.0000**
10	0.4182	0.5491	0.4790	**0.9860**	0.7215	0.5480	0.8973	0.9103	0.2379	0.6123	0.6187	0.9393
11	0.0000	0.0000	0.7444	**0.9579**	0.5511	0.0050	0.0000	0.9082	0.0000	0.0000	0.0000	0.0000
12	0.9663	0.9907	0.8267	0.0000	0.5051	0.2190	1.0000	0.4312	0.8583	0.0000	0.9808	**1.0000**
13	0.9858	**0.9902**	0.9640	0.0000	0.0656	0.0528	0.0000	0.0000	0.0000	0.0000	0.9771	0.0000
14	0.8841	0.8757	0.1159	0.0000	0.0013	0.7550	0.0000	0.0150	0.0000	0.0000	**0.8925**	0.0112
15	0.4890	0.3195	0.2558	0.8565	0.9477	0.1980	0.9747	0.9309	0.8176	0.0000	0.4241	**1.0000**
16	0.4134	0.3985	0.4438	1.0000	0.2911	0.0004	1.0000	0.9624	0.9723	0.0000	0.7559	**1.0000**
OA	0.6453	0.5631	0.5538	0.7777	0.7193	0.5417	0.7330	0.7709	0.7066	0.6152	0.7627	**0.8514**
κ	0.6430	0.5609	0.5480	0.7521	0.6900	0.4824	0.7086	0.7471	0.6741	0.5605	0.7360	**0.8350**
NMI	0.7178	0.6881	0.6618	0.8472	0.8288	0.6676	0.8744	0.8204	0.7841	0.7540	0.8151	**0.8977**
ARI	0.5118	0.4525	0.4525	0.6950	0.6278	0.4773	0.7151	0.6468	0.6096	0.5767	0.6339	**0.7669**
Purity	0.7041	0.6909	0.6750	0.7878	0.7324	0.5790	0.8349	0.7779	0.7129	0.6161	0.7869	**0.8542**
Training Time (s)	131.1816	841.0180	93.5233	681.0274	672.1873	437.5881	199.9460	144.8671	73.4302	409.8799	39.6041	**3.9806**
Testing Time (s)	----	----	----	15.6768	4.2803	0.6087	4.8399	43.5324	5.1500	0.3626	12.7679	**0.0011**
FLOPs	----	----	----	303.18 G	490.69 M	446.10 G	270.38 M	130.02 M	148.72 M	164.34 G	999.98 M	**80.75 M**

DAEGC [21], SEDESC [43], and DSCRLE [17]. Among them, *K*-means, FCM, and PCM are traditional clustering methods, NCSC and SEDESC represent subspace clustering methods, DAEGC and DSCRLE represent graph clustering method and SC method, respectively, DFCN and SDCN represent deep clustering methods, and EGAE and AdaGAE represent graph AE methods. The hyperparameters and network structure of the above methods are based on the suggestions in the original paper.

3) Parameter Settings and Implementation Details: To facilitate fair comparisons, all the experiments were conducted on a workstation using the PyTorch platform, equipped with an Intel Xeon Silver 4210R CPU, 128-GB RAM, and an NVIDIA Titan RTX 24GB GPU. Our proposed S^2GCL uses the Adam optimizer with a batch size of 64. The initial learning rate is set to 5×10^{-4} and is decayed by a factor of 0.9 after each tenth of the total epochs. We configured the number of epochs to 200 for the four datasets. It is worth noting that the MLP structure consists of two hidden layers, while the S^2GCN encoder comprises only one layer. This design aims to streamline the network structure and enhance computational efficiency.

C. Comparative Experimental Results and Analysis

This section compares S^2GCL with traditional methods and the most widely used and advanced deep learning methods. Tables II–V show the clustering accuracy of different methods, along with their running time and FLOPs. The dashed lines in the table indicate that traditional clustering algorithms do not have calculated FLOPs.

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TABLE III
QUANTITATIVE COMPARISON OF THE UP DATASET, WITH BEST PERFORMANCE IN BOLD

Class No.	K-means	FCM	PCM	NCSC	DFCN	SDCN	EAGE	AdaGAE	DAEGC	SEDESC	DSCRLE	S^2GCL
1	0.8845	0.5610	0.5538	0.4176	0.4051	0.5949	0.2811	0.8470	0.3759	**1.0000**	0.9214	0.5403
2	0.3770	0.3163	0.4633	0.5833	0.4008	0.3754	0.5117	0.5440	0.7935	0.7862	0.3909	**0.8087**
3	0.0000	0.0019	0.6027	0.6899	0.7038	0.1525	**0.9869**	0.7300	0.9609	0.0000	0.0000	0.7413
4	0.5842	0.4997	0.1648	0.1106	0.3547	0.4343	0.2650	0.7050	0.6505	**0.9354**	0.4905	0.6602
5	0.6052	0.6959	0.0000	0.9673	0.6892	0.6221	1.0000	0.0000	1.0000	0.9881	0.9948	**1.0000**
6	0.4269	0.3398	0.3265	0.9622	0.4940	0.3309	**1.0000**	0.4870	0.3327	0.0000	0.3382	0.5582
7	0.0000	**0.9684**	0.0068	0.3564	0.3486	0.3446	0.0000	0.0000	0.5301	0.0000	0.0000	0.0000
8	**0.9620**	0.8672	0.5538	0.3873	0.3444	0.4938	0.6167	0.2240	0.1645	0.0000	0.9174	0.5378
9	**0.9989**	0.0000	0.9989	0.0000	0.1683	0.2953	0.0655	0.4467	0.0847	0.0000	0.9842	0.0908
OA	0.5175	0.4273	0.4376	0.5488	0.4214	0.4127	0.5376	0.5420	0.6010	0.5958	0.5202	**0.6653**
κ	0.5165	0.4211	0.4261	0.4514	0.3189	0.3090	0.4474	0.4000	0.4866	0.4509	0.4298	**0.5670**
NMI	0.5415	0.4995	0.4758	0.4344	0.4355	0.4349	0.5249	0.4690	0.5142	0.5045	0.5550	**0.5645**
ARI	0.3117	0.2568	0.2783	0.3811	0.2691	0.2397	0.3761	0.3210	0.5049	0.3692	0.3314	**0.5546**
Purity	0.6981	0.6717	0.6487	0.6597	0.5966	0.6388	0.7078	0.6670	0.6842	0.5962	0.7104	**0.7285**
Training Time (s)	126.2800	462.3236	205.6314	1375.5680	1802.5410	274.8834	1763.6860	1200.8190	581.7761	400.4575	44.3309	**1.1341**
Testing Time (s)	----	----	----	33.2784	8.5483	0.2034	39.8889	101.3368	44.8480	0.2669	14.6631	**0.0011**
FLOPs	----	----	----	596.49 G	1.70 G	339.56 G	2.63 G	1.29 G	1.46 G	125.17 G	853.05 M	**136.97 M**

TABLE IV
QUANTITATIVE COMPARISON OF THE TRENTO DATASET, WITH BEST PERFORMANCE IN BOLD

Class No.	K-means	FCM	PCM	NCSC	DFCN	SDCN	EAGE	AdaGAE	DAEGC	SEDESC	DSCRLE	S^2GCL
1	0.0000	0.0000	0.3673	1.0000	0.6844	0.2146	0.6498	0.9950	1.0000	0.2940	0.6031	**1.0000**
2	0.0017	0.0000	0.0000	0.0575	0.4191	0.7802	0.4897	0.4795	0.1151	0.0000	**0.7775**	0.3169
3	0.0000	0.0000	0.0000	0.0000	0.0063	0.0257	**0.1581**	0.0146	0.0000	0.0000	0.0042	0.0000
4	0.0001	0.1495	0.4502	1.0000	0.9745	0.6519	0.9542	0.9528	1.0000	0.9909	0.9944	**1.0000**
5	0.6090	0.2012	0.4292	0.6354	0.7105	0.7369	0.5087	0.5808	0.6156	**0.9992**	0.7045	0.9956
6	0.0192	0.0243	0.0006	0.8135	0.3959	0.5821	0.3329	0.8100	0.8233	**0.9597**	0.7268	0.6301
OA	0.6280	0.5084	0.3678	0.7473	0.7145	0.6181	0.6362	0.7538	0.7469	0.7866	0.7768	**0.8781**
κ	0.5969	0.4887	0.3232	0.6732	0.6314	0.5067	0.5483	0.6851	0.6741	0.7060	0.7168	**0.8355**
NMI	0.4907	0.4621	0.2898	0.7276	0.6787	0.4831	0.6750	0.7212	0.6767	0.7736	0.7750	**0.7913**
ARI	0.3424	0.2811	0.1303	0.7081	0.6795	0.4320	0.6598	0.6407	0.6483	0.7547	0.7454	**0.8703**
Purity	0.6480	0.6658	0.5146	0.8631	0.7817	0.6972	0.8341	0.7666	0.8382	0.7895	0.8788	**0.8859**
Training Time (s)	119.3118	826.7424	2.9189	677.5501	282.6037	145.6749	186.1897	86.2017	65.9399	144.1920	10.6596	**0.6604**
Testing Time (s)	----	----	----	19.3583	4.4386	0.3492	4.7581	13.7605	4.7916	0.1708	7.6173	**0.0004**
FLOPs	----	----	----	273.67 G	83.27 M	236.22 G	57.09 M	25.53 M	27.75 M	86.99 G	794.42 M	**21.99 M**

TABLE V
QUANTITATIVE COMPARISON OF THE LONGKOU DATASET, WITH BEST PERFORMANCE IN BOLD

Class No.	K-means	FCM	PCM	NCSC	DFCN	SDCN	EAGE	AdaGAE	DAEGC	SEDESC	DSCRLE	S^2GCL
1	0.5782	0.5209	0.3980	0.5771	**0.8650**	0.3155	0.5907	0.6042	0.5991	0.5592	0.6537	0.8177
2	0.2944	0.3947	0.9195	0.9195	0.8914	0.0000	0.8929	0.4639	0.8199	**0.9756**	0.5147	0.8776
3	0.1947	0.3039	0.0007	0.6018	0.0494	0.0002	0.0000	0.2945	0.0000	0.0000	**0.8235**	0.0000
4	0.3153	0.2856	0.7182	0.2764	0.5628	**0.9254**	0.2958	0.6074	0.4930	0.4869	0.4334	0.4406
5	0.6398	0.0000	0.0000	0.0000	0.0000	0.0015	**0.9262**	0.4759	0.0000	0.0000	0.0000	0.0000
6	0.4894	0.4169	0.5175	0.8794	0.8409	0.6855	0.7379	0.5829	0.8857	0.7637	0.9229	**0.9647**
7	0.9998	**0.9998**	0.5023	0.4848	0.6568	0.8388	0.3853	0.4983	0.8306	0.8395	0.6335	0.9897
8	**0.5700**	0.5470	0.0014	0.2892	0.3270	0.5247	0.0862	0.2616	0.0133	0.0000	0.5345	0.1018
9	0.3861	0.4056	0.3966	0.0979	0.0793	0.0791	0.2201	0.2913	0.3396	0.0193	0.0046	**0.8329**
OA	0.6088	0.5782	0.5316	0.4518	0.6346	0.6743	0.4241	0.5363	0.6198	0.6047	0.5574	**0.7153**
κ	0.6077	0.5769	0.5258	0.3662	0.5580	0.5764	0.3441	0.4351	0.5369	0.5117	0.4821	**0.6468**
NMI	0.6140	0.5987	0.5690	0.5079	0.6245	0.6171	0.5346	0.5302	0.6331	0.5652	0.6679	**0.6873**
ARI	0.6373	0.6220	0.4604	0.5598	0.5800	0.6605	0.6492	0.5298	0.5247	0.5004	0.4944	**0.7213**
Purity	0.8024	0.7857	0.7659	0.7476	0.8321	0.8073	0.7546	0.7295	0.8155	0.7604	0.7650	**0.8597**
Training Time (s)	280.2146	8050.9018	118.8106	2780.5293	1209.3457	185.7517	333.8326	509.8570	114.0438	477.3280	86.7183	**1.2165**
Testing Time (s)	----	----	----	135.9921	8.3118	0.5945	8.5645	47.6498	8.5430	0.6975	21.2748	**0.0010**
FLOPs	----	----	----	1.38 T	1.19 G	1.73 T	1.15 G	551.68 M	629.81 M	598.55 G	1.07 G	**40.71 M**

1) Quantitative Analysis of SA Dataset: From the experimental results in Table II, S^2GCL achieves the best results among all key indicators, indicating the significant potential of the GCL method in improving the performance of HSI clustering tasks. Specifically, although traditional methods have lower OA than deep learning methods, K-means achieves the best clustering results among traditional methods. Both the graph AE methods, EGAE and AdaGAE, have achieved commendable clustering results, mainly because graph AE can improve the integrity of graph-embedded data by aggregating domain nodes. NCSC achieves the second-best OA due to its mining of implicit data structures. It is worth noting

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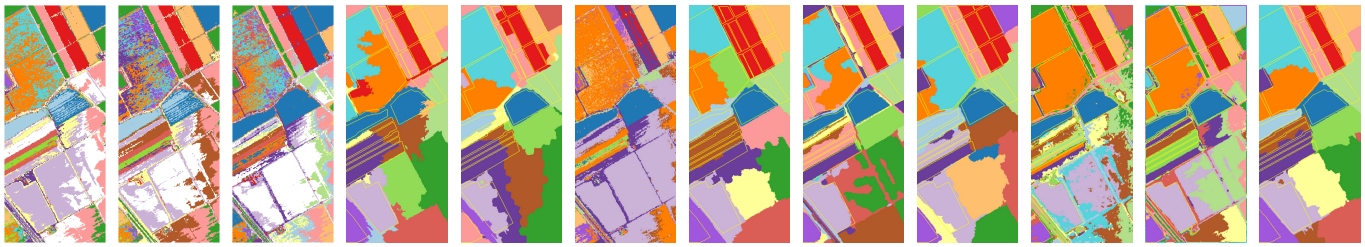


Fig. 4. Clustering maps of the SA dataset. (a) K-means. (b) FCM. (c) PCM. (d) NCSC. (e) DFCN. (f) SDCN. (g) EGAE. (h) AdaGAE. (i) DAEGC. (j) SEDESC. (k) DSCRLE. (l) S^{2}GCL.

that S^{2}GCL mines mutual information between positive and negative embeddings through GCL, autonomously learning the node features from similar data. Moreover, S^{2}GCL can focus on detailed features and clusters in the abstract-level semantic space. Compared with the best results in all the methods, S^{2}GCL has improved OA, κ, NMI, ARI, and Purity by 7.37%, 8.29%, 5.05%, 7.19%, and 6.64%, respectively, which indicates that graph contrastive learning has good application prospects in HSI clustering.

2) Quantitative Analysis of UP Dataset: Due to the relatively small size and sparse land-cover distribution, the UP dataset's clustering difficulty is higher than the SA dataset. However, S^{2}GCL still achieved outstanding performance in UP dataset clustering. Specifically, compared with the second-best clustering method, DAEGC, S^{2}GCL improved OA, κ, NMI, ARI, and Purity by 6.43%, 8.04%, 5.03%, 4.97%, and 4.43%, respectively. The DAEGC method based on graph clustering can better capture spatial graph structures, thus achieving good clustering results. However, the clustering performance of deep clustering methods DFCN and SDCN is not very satisfactory, which indicates that the feature extraction ability of deep clustering methods is insufficient when the land covers are small. S^{2}GCL learns spatial structural information through spectral–spatial embedding and neighbor embedding. In addition, global–local semantic information between different features is considered in the feature extraction process, which enables S^{2}GCL to achieve clustering accuracy far superior to other competitors.

3) Quantitative Analysis of Trento Dataset: From Table IV, we can see that our method achieved satisfactory results in all the metrics on the Trento dataset, indicating that our model has good extraction ability for the spectral–spatial features of HSI. Specifically, compared with the second best clustering method SEDESC, S^{2}GCL improved OA, κ, NMI, ARI, and Purity by 9.15%, 12.95%, 1.77%, 11.56%, and 9.64%, respectively. AdaGAE based on graph AE uses non-Euclidean geometric structures and extracts high-level semantic information more fully through adaptive weighted graphs, achieving good clustering results. In addition, NCSC and DAEGC have improved the nonlinear mapping ability in the spatial domain by establishing graph structures. Compared with the above methods, this article adopts a spectral–spatial attention mechanism while aggregating neighbor node information to generate positive embeddings, thereby achieving joint feature extraction of spatial and spectral data. The experimental results also show that this method is essential in improving clustering performance.

4) Quantitative Analysis of LongKou Dataset: From the quantitative clustering results in Table V, the following conclusions can be drawn. First, compared with the SA and Trento datasets, the clustering accuracy of all the methods on the LongKou dataset is not high. Due to the similarity in the types of land covers and their close spectral characteristics of the LongKou dataset, misclassification is prone to occur, which poses a greater challenge for the investigated algorithm. Second, deep clustering methods such as DFCN and SDCN outperform shallow methods such as K-means, indicating that deep clustering methods can extract higher level semantic relationships and better integrate spectral–spatial features to establish relationships between the original data, exhibiting greater adaptability. Furthermore, S^{2}GCL still achieves the best clustering performance, with improvements of 4.1%, 7.04%, 7.02%, 6.08%, and 5.24% in OA, κ, NMI, ARI, and Purity, respectively. This is attributed to the design of the spatial and spectral pixel matrix, which facilitates better spectral differentiation of various land-cover features, reduces misclassification, and demonstrates good adaptability across different datasets.

D. Comparison of Runtime and Model Complexity

To compare the computational cost and model complexity of different methods, Tables II–V presents the training and testing times on four datasets and shows comparison of the model complexity of different methods through FLOPs. The lightweight graph contrastive learning network S^{2}GCL designed in this article eliminates the data augmentation and identification modules that require a large number of computing resources, significantly saving computation time and reducing model complexity. The results show that our method's training and testing times and model complexity are much lower than other methods, and the quantitative experiments also show that S^{2}GCL achieves remarkable clustering results while reducing computational costs. Therefore, the more minor model complexity and higher clustering accuracy of S^{2}GCL make it have broad industrial application prospects.

E. Visual Clustering Effect Analysis

To visually compare the clustering results of different methods, this article conducts qualitative evaluations of different

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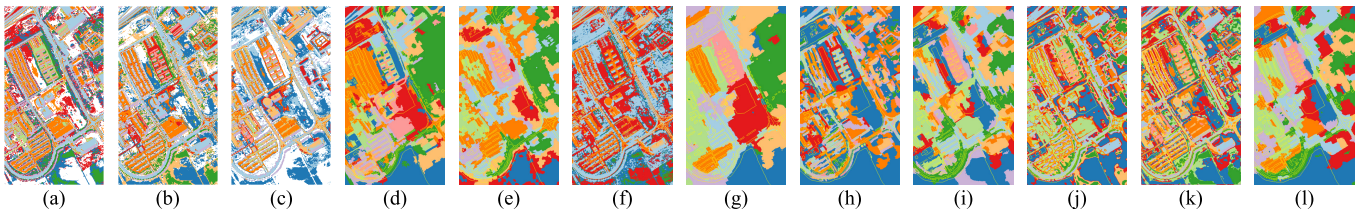


Fig. 5. Clustering maps of the UP dataset. (a) K-means. (b) FCM. (c) PCM. (d) NCSC. (e) DFCN. (f) SDCN. (g) EGAE. (h) AdaGAE. (i) DAEGC. (j) SEDESC. (k) DSCRLE. (l) S^{2}GCL.

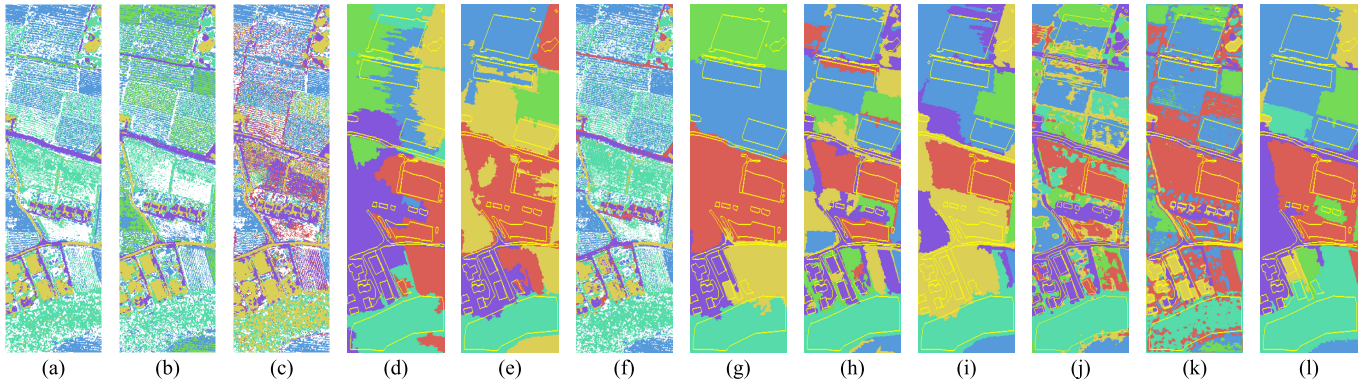


Fig. 6. Clustering maps of the Trento dataset. (a) K-means. (b) FCM. (c) PCM. (d) NCSC. (e) DFCN. (f) SDCN. (g) EGAE. (h) AdaGAE. (i) DAEGC. (j) SEDESC. (k) DSCRLE. (l) S^{2}GCL.

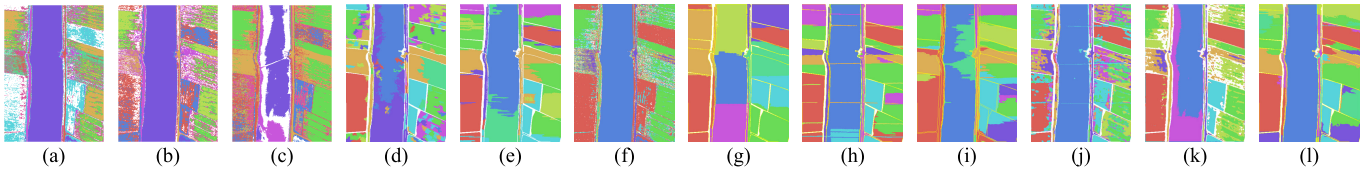


Fig. 7. Clustering maps of the LongKou dataset. (a) K-means. (b) FCM. (c) PCM. (d) NCSC. (e) DFCN. (f) SDCN. (g) EGAE. (h) AdaGAE. (i) DAEGC. (j) SEDESC. (k) DSCRLE. (l) S^{2}GCL.

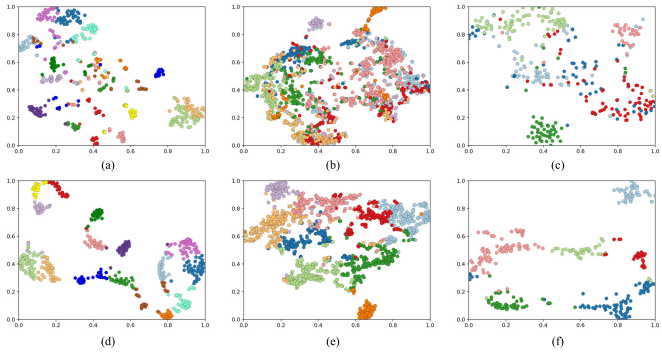


Fig. 8. t-SNE visualization of S^{2}GCL on (a) and (d) SA, (b) and (e) UP, and (c) and (f) Trento. (a)–(c) Original distribution of three datasets. (d)–(f) Data distribution after clustering. Different colors in the maps denote different land covers.

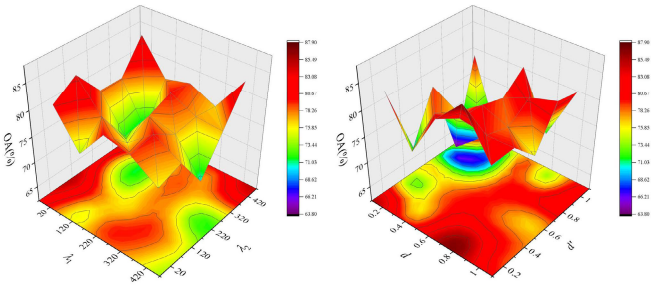


Fig. 9. Sensitivity of S^{2}GCL in different parameter settings (i.e., λ_1 and λ_2, d and d') on the Trento dataset.

methods through visual experiments. Figs. 4–7 show the clustering results of various methods on four datasets, where we use yellow solid lines to indicate the boundaries of the land covers for better observation. Figs. 4–7 illustrate that traditional clustering methods (K-means, FCM, and PCM) are prone to generating salt and pepper noise, indicating that their extraction of land features is insufficient and cannot effectively cluster. Due to its nonlinear solid feature fitting ability, deep learning methods have obtained smoother and more accurate clustering maps. As a novel exploration of graph contrastive learning in HSI clustering, S^{2}GCL has achieved impressive clustering results while significantly reducing model complexity. We use the S^{2}GCN encoder to aggregate neighbor information through the spatial structure matrix, which better perceives the local spatial information of the land cover. In addition, S^{2}GCL focuses on the texture and detail features of the terrain, reducing misclassification and making the

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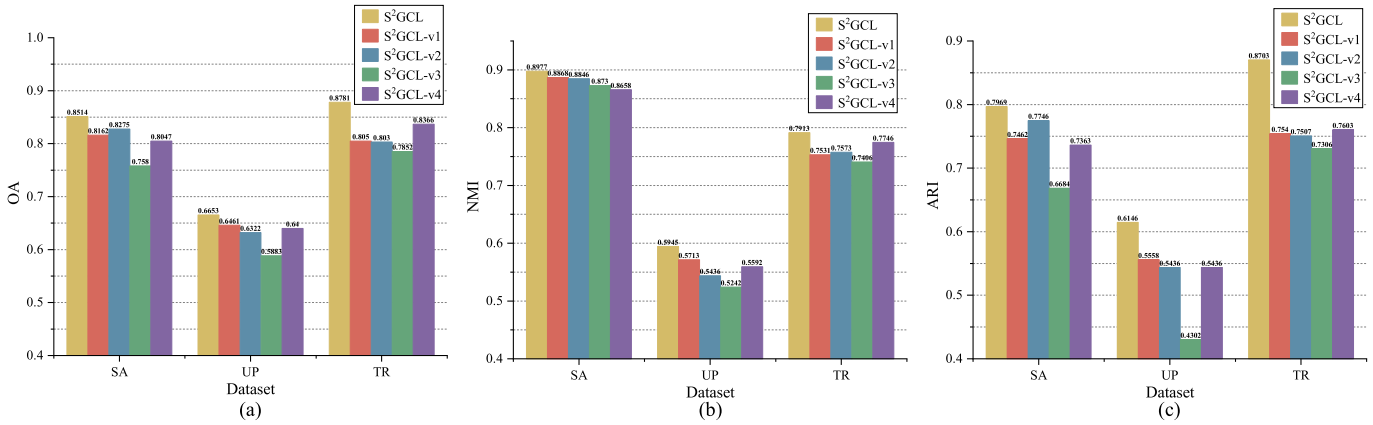


Fig. 10. Ablation study results on SA, UP, and Trento. (a) OA, (b) NMI, and (c) ARI are reported.

clustering effect smoother. In addition, the multiple boundary loss proposed in this article can better constrain positive and negative embeddings, enabling the network to extract advanced semantic information autonomously. Visual experimental results show that our method is effective.

F. Clustering Visualization

In this section, we visualized the distribution of graph nodes constructed from three datasets using t-distributed stochastic neighbor embedding (t-SNE) [44] to validate the clustering performance of S^{2}GCL. As shown in Fig. 8,(a)–(c) represents the original distribution of graph nodes, while Fig. 8(d)–(f) represents the distribution of graph node features after encoding with S^{2}GCL. Based on the colormap analysis, the following conclusions can be drawn: 1) compared with the original distribution, the covariance between different classes in the encoded feature distribution is smaller, indicating weaker variable correlations, which is more conducive to clustering; 2) the probability of misclassification is reduced; and 3) nodes of the same class exhibit smaller intraclass distances and larger interclass distances, resulting in a more regular and compact distribution. In summary, the features encoded by S^{2}GCL exhibit tighter intraclass compactness, better interclass separation, and effectively enhance clustering performance.

G. Parameter Study

In this section, we investigated the influence of hyperparameters on the experimental results. We used a grid search strategy to find the optimal settings for loss function weights λ_1 and λ_2, as well as boundary distances d and d'. As shown in Fig. 9, we present the results of the OA values of two sets of parameters on the Trento dataset, with other settings in the network being the same as in the above experiments. The experimental results indicate that the OA is highest when d and d' take 0.8 and 0.2, respectively. Increasing the distances d and d' appropriately improves clustering accuracy, as it increases the distance between positive and negative pairs. When λ_1 and λ_2 take 20 and 120, OA is the highest, and OA is very sensitive to the changes in weights λ_1 and λ_2, indicating that both $\mathcal{L}_G$ and $\mathcal{L}_N$ are crucial for the effectiveness of feature extraction.

TABLE VI
ABLATION EXPERIMENTAL SETUP IN S^{2}GCL

Name		S^{2}GCL	S^{2}GCL -v1	S^{2}GCL -v2	S^{2}GCL -v3	S^{2}GCL -v4
Module	Spatial Structure Matrix	✔	✘	✔	✔	✔
	Spectral Similarity Matrix	✔	✔	✘	✔	✔
	S^{2}GCN Encoder	✔	✔	✔	✘	✔
	Neighbor Encoder	✔	✔	✔	✔	✘

H. Ablation Experiment

In this section, we designed ablation experiments to evaluate the contributions of the spatial structure matrix, spectral similarity matrix, S^{2}GCN encoder, and lightweight neighbor encoder within S^{2}GCL by assessing their impact on clustering accuracy on the SA, UP, and Trento datasets. The experimental setup remains consistent with that described in Section IV-B3. Clustering accuracy was reported using OA, NMI, and ARI metrics, with the results presented in Fig. 10. To facilitate a straightforward and intuitive validation, we conducted experiments by sequentially removing the spatial structure matrix, spectral similarity matrix, S^{2}GCN encoder, and lightweight neighbor encoder from S^{2}GCL, denoting the reduced methods as "S^{2}GCL-v1" to "S^{2}GCL-v4," with the configurations detailed in Table VI.

Fig. 10 indicates that S^{2}GCL achieves the best performance across all three metrics. The decline in clustering performance in "S^{2}GCL-v1" to "S^{2}GCL-v4" due to the absence of certain modules indicates that each module contributes varying degrees of improvement to the clustering accuracy of S^{2}GCL. Specifically, the substantial performance drop in S^{2}GCL-v3 highlights the importance of the S^{2}GCN encoder, which effectively aggregates spatial and spectral information to enhance feature extraction and improve clustering results. Furthermore, the declines in clustering accuracy for S^{2}GCL-v1 and S^{2}GCL-v2 indicate that using the spatial structure matrix and spectral similarity matrix to capture spatial relationships and establish a local spectral attention mechanism effectively enhances clustering performance. The results of S^{2}GCL-v4 demonstrate that using a lightweight neighbor

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encoder to represent positive samples can further enhance S^{2}GCL's contrastive effectiveness. Therefore, the spatial structure matrix, spectral similarity matrix, S^{2}GCN encoder, and lightweight neighbor encoder within S^{2}GCL have collectively contributed to improved clustering performance.

V. Conclusion

This article presents an efficient and lightweight graph contrastive learning network, S^{2}GCL, designed for extracting deep spectral–spatial semantic information from HSI. Specifically, we eliminate traditional graph augmentation and graph discriminators in contrastive learning, which significantly enhances model efficiency and reduces complexity. Furthermore, the S^{2}GCN encoder is developed to generate spectral–spatial structural embeddings, facilitating the joint extraction of spectral and spatial information by incorporating both a spatial structure matrix and a spectral similarity matrix. Subsequently, to enhance the utilization of neighbor information, we designed the neighbor embeddings to extract local spatial semantic features. To streamline the model structure, a row-shuffling module was implemented to efficiently generate negative embeddings. Finally, we developed a multiple boundary loss to regulate the relationship between positive and negative embeddings, ensuring consistency and uniformity among the embedding pairs.

The experimental results demonstrate that S^{2}GCL excels in jointly extracting spectral–spatial features, achieving the highest clustering accuracy compared with other clustering methods. Furthermore, S^{2}GCL is relatively lightweight and efficient, characterized by low model complexity and the shortest running time, which significantly enhances computational efficiency and underscores its promising application prospects. In the future, we will further investigate methods for HSI preprocessing to enable finer-grained feature extraction, as well as explore additional clustering methods based on graph contrastive learning to achieve even higher clustering accuracy.

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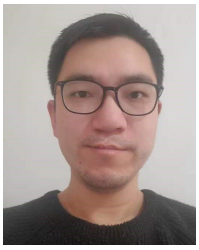
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